

UE19MA251 - LINEAR ALGEBRA AND ITS APPLICATIONS

QUESTION BANK

UNIT 4 – ORTHOGONALIZATION. EIGEN VALUES AND EIGEN VECTORS

1	The set $\{v_1, v_2, v_3\}$ is an orthogonal basis in R^3. Express the vector $y = \begin{pmatrix} 6 \\ 1 \\ -8 \end{pmatrix}$ as a linear combination of the vector in S, where $v_1 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$ $v_2 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$ $v_3 = \begin{pmatrix} -\frac{1}{2} \\ -2 \\ \frac{7}{2} \end{pmatrix}$ ANS: $y = v_1 - 2v_2 - 3v_3$
2	Find the distance from the point $z = (0, 0, 1, 0)$ to the plane P that passes through the point $x_0 = (1, 0, 0, 0)$ and is parallel to the vectors $v_1 = (1, -1, 1, -1)$ and $v_2 = (0, 2, 2, 0)$. Answer: $\text{Sqrt}(3)/2$
3	Let P be the plane in R^3 spanned by vectors $x_1 = (1, 2, 2)$ and $x_2 = (-1, 0, 2)$. i) Find an Orthogonal basis for P ii) Extend it to an Orthonormal basis for R^3 Answer: i) $\frac{1}{3}(1, 2, 2), \frac{1}{3}(-2, -1, 2)$ ii) $\frac{1}{3}(1, 2, 2), \frac{1}{3}(-2, -1, 2), \frac{1}{3}(2, -2, 1)$
4	Perform the Gram-Schmidt process on the following sets and also give the associated QR factorization. i) $a = (1, 2, 2), b = (1, 3, 1)$ ii) $a = (2, 2, 1), b = (1, 1, 5)$ iii) $a = (2, 2, 1), b = (-2, 1, 2), c = (18, 0, 0)$

	Answer : (i) $q_1 = \frac{1}{3} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, $q_2 = (1/\sqrt{2}) \cdot (0, 1, -1)$ and $\begin{bmatrix} 1/3 & 0 \\ 2/3 & 1/\sqrt{2} \\ 2/3 & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 0 & \sqrt{2} \end{bmatrix}$ (ii) $q_1 = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$, $q_2 = \frac{1}{\sqrt{18}} \begin{bmatrix} -1 \\ -1 \\ -4 \end{bmatrix}$ and $\begin{bmatrix} 2/3 & -1/\sqrt{18} \\ 2/3 & -1/\sqrt{18} \\ 1/3 & 4/\sqrt{18} \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 0 & \sqrt{18} \end{bmatrix}$ (iii) $q_1 = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$, $q_2 = \frac{1}{3} \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$, $q_3 = \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$ and $\frac{1}{3} \begin{bmatrix} 2 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 & 12 \\ 0 & 3 & -12 \\ 0 & 0 & 6 \end{bmatrix}$
5	What multiple of $a_1 = (1, 1)$ should be subtracted from $a_2 = (4, 0)$ to make the result orthogonal to a_1? Factor the matrix in to QR with Orthonormal vectors in Q. Answer : multiple 2, $A = QR = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & 2\sqrt{2} \end{bmatrix}$
6	Determine the Eigen values and corresponding Eigen vectors for the following matrices: (i) $\begin{bmatrix} -5 & 3 & 0 \\ -6 & 4 & 2 \\ 2 & -1 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$ Answer: (i) -1, 0, 1 and $k_1(-3, -4, 1)$, $k_2(-3, -5, 1)$, $k_3(0.5, 1, 0)$ where k_1, k_2, k_3 are real (ii) 2, -2, -2 and $x_3(0.5, 0.5, 1)$, $x_3(1, 0, 1) + x_2(-1, 1, 0)$, where x_1, x_2, x_3 are real
7	If A has $\lambda_1 = 1$ with Eigen vector $X_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $\lambda_2 = 5$ with Eigen vector $X_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, find A.
8	Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ $u = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$, $v = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$. Are u and v eigenvalues of A ? Answer : u but not v is an eigenvalue of A.
9	Find the eigen values and the corresponding eigenvectors of $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$ Ans: The eigenvectors for $\lambda = 2$ are $a(1, 2, 0)$ and $b(-3, 0, 1)$ The eigenvector for $\lambda = 9$ are $c(1, 1, 1)$ where a, b, c are nonzero scalars

10	If possible diagonalize the matrix $A = \begin{bmatrix} 1 & 1 \\ -1 & 4 \end{bmatrix}$ Answer: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$
11	Diagonalize the following matrices: $\begin{bmatrix} 3 & 0 & 0 \\ -3 & 4 & 9 \\ 0 & 0 & 3 \end{bmatrix}, \begin{bmatrix} -4 & -4 & -8 \\ 4 & 6 & 4 \\ 6 & 4 & 10 \end{bmatrix}$ Answer: $\begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & -3 \\ -3 & 1 & 9 \end{bmatrix}$ $\begin{bmatrix} -2 & -1 & -2 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} -1 & 0 & -1 \\ -1 & -2 & 0 \\ 1 & 1 & 1 \end{bmatrix}$
12	Check whether the following symmetric matrices are Diagonalizable (i) $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}$
13	Find a formula for $\begin{bmatrix} 1 & -6 \\ 2 & -6 \end{bmatrix}^k$ by diagonalizing the matrix. Answer: $A^k = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} (-3)^k & 0 \\ 0 & (-2)^k \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix}$
14	Calculate A^6, where $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$, Answer: $A^6 = \begin{bmatrix} -600 & 665 \\ -1330 & 1394 \end{bmatrix} = P D^6 P^{-1}$
15	Show that $A = \begin{bmatrix} 1 & 7 & 6 \\ 0 & -1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$ is diagonalizable.

